

MINGI KANG

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EDUCATION

Bowdoin College, ME

B.A. in Computer Science, Minor in Mathematics
Overall GPA: 3.71 / 4.00

August 2022 - May 2026

Aquincum Institute of Technology, Budapest, Hungary

Study Abroad Semester, Computer Science

August - December 2024

PUBLICATIONS, PRESENTATIONS, MEDIA

Publications

- [1] FCT: Fully Convolutional Transformers with Linear Attention *In Preparation*
[Mingi Kang](#), [Jeová Farias Sales Rocha Neto](#) 2026
- [2] IGLU: The Integrated Gaussian Linear Unit Activation Function *Under Review*
[Mingi Kang](#), [Zai Yang](#), [Jeová Farias Sales Rocha Neto](#) 2026
[arXiv preprint arXiv:2603.06861](#)
- [3] Attention Via Convolutional Nearest Neighbors *Preprint*
[Mingi Kang](#), [Jeová Farias Sales Rocha Neto](#) 2025
[arXiv preprint arXiv:2511.14137](#)
- [4] Parallel qMRI Reconstruction from 4x Accelerated Acquisitions *Preprint*
[Mingi Kang](#) 2025
[arXiv preprint arXiv:2511.18232](#)
- [5] Structures and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing *Conference Publication*
[Abhilasha A. Kumar](#), [Mingi Kang](#), [William G. Kronenberger](#), [Michael N. Jones](#), [David B. Pisoni](#) 2024
CogSci 2024. Proceedings of the 46th Annual Meeting of the Cognitive Science Society (Vol. 46).
DOI: <https://escholarship.org/uc/item/7vn9q9hh>

Presentations

- [1] Parallel qMRI Reconstruction from 4x Accelerated Acquisitions *Oral*
[Mingi Kang](#) July 2025
McKelvey School of Engineering Summer Symposium, St. Louis, MO
- [2] Convolutional Nearest Neighbors: Reinterpreting Convolution Through K-Nearest Neighbor Selection *Poster*
[Mingi Kang](#) October 2025
IEEE MIT Undergraduate Research Technology Conference (MIT URTC), Cambridge, MA
- [3] Parallel qMRI Reconstruction from 4x Accelerated Acquisitions *Poster*
[Mingi Kang](#) July 2025
McKelvey School of Engineering Poster Palooza, St. Louis
- [4] Structures and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing *Poster*
[Mingi Kang](#) July 2024
Annual Conference of the Cognitive Science Society 2024, Rotterdam, Netherlands

Media

- [1] Mingi Kang '26: Advancing Computers' Ability to See and Understand Our World *School Article*
[Bowdoin Article Link](#) November 2025

RESEARCH EXPERIENCE

Independent Deep Learning Research Group*Undergraduate Research Assistant (Advisor: Prof Jeova Farias)*

August 2025 - Present

Brunswick, ME

- Developed **IGLU** activation function that generalizes ReLU and GELU with learnable parameter σ for fine-grained control.
- Implemented modular GPT₂ architecture allowing plug-and-play activation function swapping, training on WikiText-103 to evaluate performance via perplexity metrics.
- Demonstrated 1-2% improvements on image classification on CIFAR-10/100 with VGG19 and ResNet34 compared to baseline ReLU, GELU, and SiLU activations, with consistent gains across architectures.

Senior Honors Thesis in Computer Science*Honors Candidate (Advisor: Prof Jeova Farias)*

August 2025 - Present

Brunswick, ME

- Finalizing **ConvNN**, a unified framework bridging convolution and attention through k -nearest neighbor selection, with first-author paper submitted and currently under review.
- Architected hybrid branching layers combining spatial (Conv2d) and feature-based (ConvNN) aggregation, achieving 4-8% accuracy improvements on CIFAR-10/100 with VGG-11 architecture.
- Demonstrated effective and robust performance across CNN and Transformer architectures through systematic ablation studies on neighbor selection strategies.

Computational Imaging Group, Washington University in St. Louis*McKelvey Summer Engineering Fellow (Advisor: Prof Ulugbek S. Kamilov)*

May - August 2025

St. Louis, MO

- Redesigned deep unfolding U-Net architecture for parallel qMRI reconstruction, achieving 4x parameter reduction while maintaining reconstruction quality (37 dB PSNR, 0.923 SSIM) on 4x accelerated scans.
- Developed normalization techniques (ACS region-specific and coil-instance) for undersampled k -space preprocessing, validated on patient dataset from WashU Medical School.

Independent Deep Learning Research Group*Undergraduate Research Assistant (Advisor: Prof Jeova Farias)*

January - May 2025

Brunswick, ME

- Developed **Convolutional Nearest Neighbor Attention (ConvNN-Attention)**, an efficient attention mechanism using hard k -NN and convolutional aggregation, reducing computational cost by 18% (GFLOPS) with accuracy improvement (2.4%) on CIFAR-10/100.
- Implemented modular PyTorch layers compatible with standard Transformer and Vision Transformer architectures, enabling drop-in replacement for self-attention.

Christenfeld Summer Research Fellowship*Research Fellow (Advisor: Prof Jeova Farias)*

January - August 2024

Brunswick, ME

- Developed **Convolutional Nearest Neighbors (ConvNN)** algorithm integrating norm-based k -NN pixel selection into standard convolutional operations for spatial feature learning.
- Built modular PyTorch implementation (1D/2D) with configurable sampling strategies (random, spatial), pixel-shuffling, and positional encoding.

Lexicon Lab, Bowdoin College*Undergraduate Research Assistant (Advisor: Prof Abhilasha Kumar)*

December 2022 - May 2024

Brunswick, ME

- Investigated lexical retrieval processes in prelingually deaf cochlear implant users through computational cognitive modeling, contributing to published CogSci 2024 conference proceedings paper (second author).
- Extended Python package, *Forager* with joint semantic embeddings (word2vec, speech2vec) for quantitative analysis, revealing differential reliance on speech-derived representations.

TEACHING/MENTORING EXPERIENCE**Bowdoin College Baldwin Center for Learning and Teaching***Quantitative Tutor*

August 2025 - Present

Brunswick, ME

- Provide one-on-one tutoring in computer science (algorithms, data structures, AI), mathematics (linear algebra, probability, statistics), and economics (microeconomics, macroeconomics).
- Support students in debugging code, understanding theoretical concepts, and developing problem solving strategies for quantitative coursework.

Bowdoin College Mathematics Department*Learning Assistant (Teaching Assistant) for Statistics and Data Science*

August 2025 - Present

Brunswick, ME

- Conduct weekly learning assistant hours supporting 20+ students with R programming, statistical analysis, hypothesis testing, data visualization, and data cleaning.
- Guide students through applied projects involving real-world datasets, emphasizing statistical thinking and reproducible analysis.

Bowdoin College Computer Science Department*Learning Assistant (Teaching Assistant) for Intro to Computer Science*

January 2025 - Present

Brunswick, ME

- Lead weekly learning assistant hours for 40+ students covering Python fundamentals and object-oriented programming.
- Mentor students on debugging techniques, code organization, and developer best practices.

Bowdoin College Career Exploration and Development Group*Sophomore Bootcamp Leader & Pod Leader*

January 2025, January 2026

Brunswick, ME

- Design and deliver workshops on career development for 400+ sophomores, covering resume, writing, behavioral interviews, internship navigation, personal finance, academic research, graduate school, and fellowships.
- Train and manage a team of 7 senior mentors, overseeing their preparation to deliver specialized workshops and career curricula to the sophomore class.
- Mentored 5 student teams in inaugural Sophomore Bootcamp Hackathon, providing technical guidance on full-stack development, API integration, and project management.

RELEVANT COURSEWORK

Computer Science: Deep Learning for Computer Vision, Operating Systems, Artificial Intelligence, Computer Systems, Algorithms, Data Structures, Data Science, Cryptography, Computational Game Theory

Mathematics: Linear Algebra, Multivariable Calculus, Probability, Statistics, Mathematical Reasoning, Advanced Probability and Statistics

TECHNICAL SKILLS

Programming Languages	Python, R, Java, C, SQL, JavaScript, Kotlin
ML & DL	PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, SciPy, OpenCV
Visualization	Matplotlib, Seaborn, ggplot2
Developer Tools	Git, GitHub, Bash, Overleaf/LaTeX, Markdown
HPC	Slurm (sbatch), IBM LSF (bsub)
Languages	English (Native), Korean (Native), German (Conversational)

AWARDS AND HONORS

1. CRA Outstanding Undergraduate Researcher Award, Honorable Mention
Computing Research Association
December 2025
- Prestigious honor recognizes undergraduate students at North American colleges and universities who demonstrate exceptional potential in computing research. [Website Link](#)
2. John L. Roberts Fall Research Award (\$ 2,463)
Bowdoin College
October 2025
- Awarded for senior honors project on Convolutional Nearest Neighbors & Convolutional Nearest Neighbors Attention
3. Best Poster Presentation Award from McKelvey Engineering Summer Research (\$ 100)
Washington University in St. Louis
July 2025
- Awarded for excellence in summer research presentation.
4. McKelvey Engineering Summer Research Fellowship (\$ 7,200)
Washington University in St. Louis
May 2025
- Awarded to conduct 10-week full time research with the Computational Imaging Group.
5. Allen B. Tucker Computer Science Research Prize (\$ 50)
Bowdoin College
May 2025
- Awarded to a computer science student for excellence in summer research.

6. CRA Undergraduate Award (\$ 4,000) *Local*
Last Mile Education April 2025
- Awarded for supplemental assistance for summer research in computer science at Washington University in St. Louis.
7. NYC Stem Award (\$ 1,500) *Local*
Last Mile Education March 2025
- Awarded merit-based grant for technical equipment and research support.
8. Google AI 2024 Award (\$ 595) *Local*
Last Mile Education January 2025
- Awarded for semester research, building on prior work from Bowdoin College Christenfeld Summer Research Fellowship.
9. Christenfeld Summer Research Fellowship (\$ 4,800) *Institutional*
Bowdoin College April 2024
- Awarded to conduct 8-week full time research in computer vision and deep learning under faculty mentorship.
10. NSF Student Faculty Research Fellowship (\$ 1,800) *National/Institutional*
Bowdoin College, National Science Foundation January 2024
- Awarded for computational cognitive science research focused on modeling search and retrieval within the mental lexicon.
11. QuestBridge National College Match Finalist *National*
QuestBridge September 2021
- Recognized as a finalist for the QuestBridge program, which connects high-achieving students from low-income, first-generation backgrounds with full scholarships to top colleges.